RBD_Analysis

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Randomized Complete Block Design (RBD / RCBD)

A **Randomized Complete Block Design (RBD)** is used when the experimental units are *not* homogeneous, but can be grouped into blocks of similar units (for example, along a fertility

or moisture gradient in the field).

Every treatment appears **once in every block**, and the treatments are randomized separately within each block.

In this example, six fertilizer treatments are compared based on crop yield in five blocks.

The experimental structure is:

- **Experimental unit:** plot
- **Treatment:** fertilizer treatment (6 levels)
- **Block:** field block (5 blocks)
- **Response variable:** yield (t/ha)
- **Replications:** 5 (one plot per treatment in each block)
- **Total plots:** $6 \times 5 = 30$
- **Design:** RBD

The main question is:

> **Do the fertilizer treatments produce different mean yields, after removing the variation between blocks?**

---

1. Data

Load the data

```
datarbd <- read.csv("rbd_normal_data.csv")

head(datarbd)
```

```
    block   treatment yield_t_ha
1 Block_1     Control       3.29
2 Block_1         NPK       4.01
3 Block_1     Organic       3.71
4 Block_1 NPK_Organic       4.61
5 Block_1        Urea       3.80
6 Block_1     Compost       3.90
```

```
str(datarbd)
```

```
'data.frame':   30 obs. of  3 variables:
 $ block     : chr  "Block_1" "Block_1" "Block_1" "Block_1" ...
 $ treatment : chr  "Control" "NPK" "Organic" "NPK_Organic" ...
 $ yield_t_ha: num  3.29 4.01 3.71 4.61 3.8 3.9 3.6 4.3 3.75 4.68 ...
```

Check whether there are missing values:

```
colSums(is.na(datarbd))
```

```
     block  treatment yield_t_ha
         0          0          0
```

Convert block and treatment to factors

This step is important. If `block` is coded as 1, 2, 3, 4, 5, R will treat it as a number and fit a slope instead of a block effect.

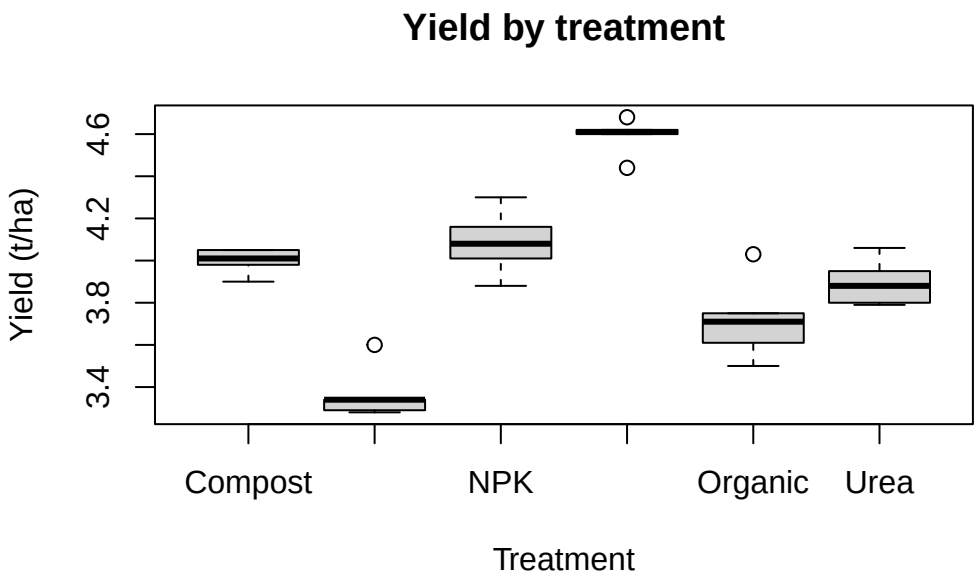
```
datarbd$block     <- as.factor(datarbd$block)
datarbd$treatment <- as.factor(datarbd$treatment)

str(datarbd)
```

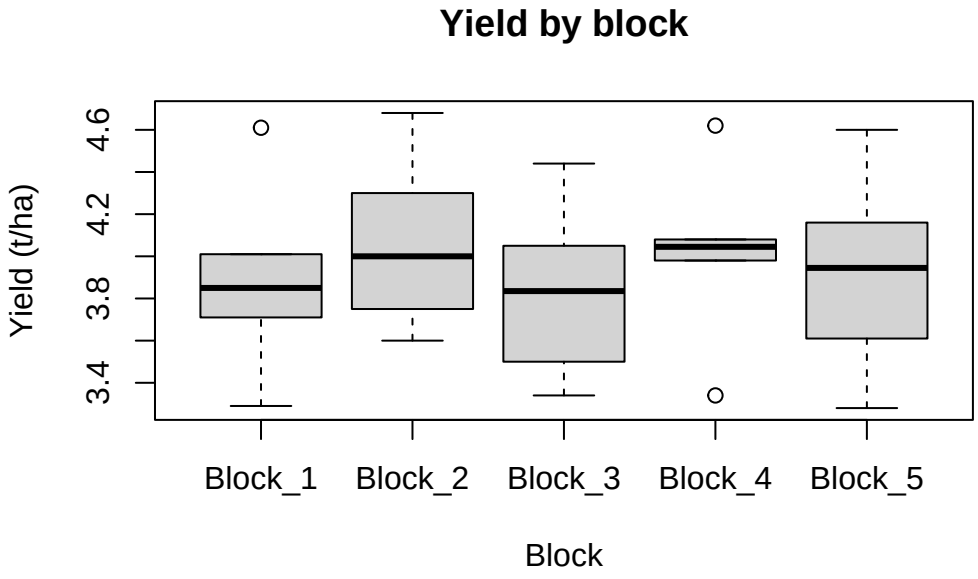
```
'data.frame':   30 obs. of  3 variables:
 $ block     : Factor w/ 5 levels "Block_1","Block_2",..: 1 1 1 1 1 1 2 2 2 2 ...
 $ treatment : Factor w/ 6 levels "Compost","Control",..: 2 3 5 4 6 1 2 3 5 4 ...
 $ yield_t_ha: num  3.29 4.01 3.71 4.61 3.8 3.9 3.6 4.3 3.75 4.68 ...
```

Quick look at the data

```
boxplot(yield_t_ha ~ treatment, data = datarbd,
        xlab = "Treatment", ylab = "Yield (t/ha)",
        main = "Yield by treatment")
```




```
boxplot(yield_t_ha ~ block, data = datarbd,
        xlab = "Block", ylab = "Yield (t/ha)",
        main = "Yield by block")
```



If the block means differ clearly, blocking was useful.

---

2. Model Building

The RBD model is:

$$Y_{ij} = \mu + \tau_i + \beta_j + \epsilon_{ij}$$

where:

- τ_i = effect of treatment i
- β_j = effect of block j
- ϵ_{ij} = random error

There is **no treatment × block interaction** in this model. The model assumes that treatments behave the same way in every block (additivity).

Fit the ANOVA model

```
modelrbd <- aov(
  yield_t_ha ~ block + treatment,
  data = datarbd
)

modelrbd
```

```
Call:
   aov(formula = yield_t_ha ~ block + treatment, data = datarbd)

Terms:
                    block treatment Residuals
Sum of Squares  0.202833  4.111747  0.221487
Deg. of Freedom        4         5        20

Residual standard error: 0.1052347
Estimated effects may be unbalanced
```

View the ANOVA table:

```
summary(modelrbd)
```

```
            Df Sum Sq Mean Sq F value   Pr(>F)
block        4  0.203  0.0507   4.579  0.00868 **
treatment    5  4.112  0.8223  74.257 3.22e-12 ***
Residuals   20  0.221  0.0111
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

At this point, we have **fitted the model**, but we should not interpret the treatment p-value yet. First, check the assumptions.

---

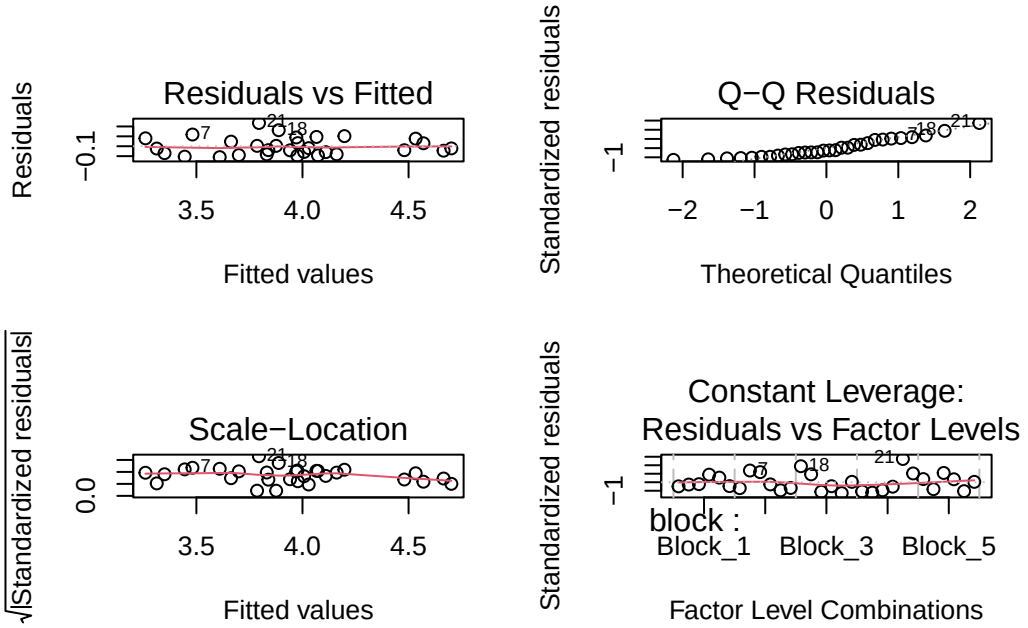
3. Model Diagnostics

For this model, the main assumptions are:

1. Independence of observations
2. Approximately normal residuals
3. Homogeneity of residual variance
4. **Additivity** (no block × treatment interaction)

3.1 Diagnostic plots

```
par(mfrow = c(2, 2))
plot(modelrbd)
```




```
par(mfrow = c(1, 1))
```

- **Residuals vs Fitted:** look for a random spread with no clear pattern.
- **Normal Q-Q:** points should approximately follow the reference line.
- **Scale-Location:** helps assess homogeneity of variance.
- **Residuals vs Leverage:** helps identify unusually influential observations.

3.2 Normality of residuals

```
shapiro.test(residuals(modelrbd))
```

```
    Shapiro-Wilk normality test

data:  residuals(modelrbd)
W = 0.92455, p-value = 0.03526
```

Interpretation:

- **$p > 0.05$:** no evidence against normality; the ANOVA is reliable.

- **$p < 0.05$:** residuals are not normal; reconsider the analysis.

Do not rely on Shapiro-Wilk alone. With small samples it has low power, and with large samples it flags trivial departures. Always read it together with the Q-Q plot.

3.3 Homogeneity of variance

Levene's test across treatments:

```
library(car)
```

```
Warning: package 'car' was built under R version 4.4.3
```

```
Loading required package: carData
```

```
leveneTest(yield_t_ha ~ treatment, data = datarbd)
```

```
Levene's Test for Homogeneity of Variance (center = median)
      Df F value Pr(>F)
group  5  0.7619 0.5861
      24
```

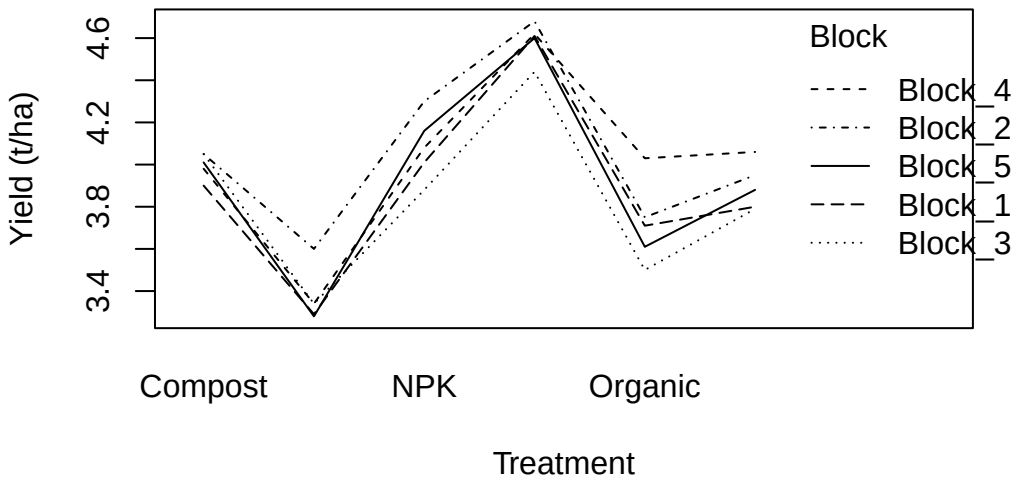
- **$p > 0.05$:** no strong evidence of unequal variances.
- **$p < 0.05$:** evidence that treatment variances differ.

With only 5 observations per treatment, this test has limited power, so the residuals vs fitted plot matters more.

3.4 Additivity (no block × treatment interaction)

The RBD has only one observation per treatment per block, so the interaction cannot be tested directly from the ANOVA table. A simple visual check is an interaction plot:

```
interaction.plot(
  x.factor = datarbd$treatment,
  trace.factor = datarbd$block,
  response = datarbd$yield_t_ha,
  xlab = "Treatment", ylab = "Yield (t/ha)",
  trace.label = "Block"
)
```



Interpretation:

- Roughly **parallel lines** → additivity is reasonable.
- Lines that **cross strongly** → treatments respond differently in different blocks, and the additive RBD model may be inadequate.

3.5 Diagnostic decision

Decision: The plots and tests show no serious violations, proceed with the RBD ANOVA.

If the assumptions were seriously violated, reconsider the analysis rather than blindly interpreting the ANOVA. For a non-normal response in an RBD, the **Friedman test** is a non-parametric alternative.

---

4. ANOVA Interpretation

```
summary(modelrbd)
```

```
            Df Sum Sq Mean Sq F value   Pr(>F)
block        4  0.203  0.0507   4.579  0.00868 **
treatment    5  4.112  0.8223  74.257 3.22e-12 ***
Residuals   20  0.221  0.0111
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Treatment effect

- **Null hypothesis (H):** all treatment means are equal.
- **Alternative hypothesis (H):** at least one treatment mean differs.

Use the significance level:

$$\alpha = 0.05$$

Decision rule:

- **p $<$ 0.05:** reject H .
- **p 0.05:** fail to reject H .

Decision: reject H for treatment, As the p value is less than o.o5 means the treatment are significantly different.

Block effect

The block also different. It is **not** a hypothesis of primary interest, but it tells us whether blocking was worthwhile.

4.1 Was blocking efficient?

The **relative efficiency (RE)** compares the RBD with what a CRD would have given for the same data:

$$RE = \frac{(b-1)MS_{Block} + b(t-1)MS_{Error}}{(bt-1)\,MS_{Error}}$$

```r
aov_tab <- summary(modelrbd)[[1]]

MS_block <- aov_tab["block", "Mean Sq"]
MS_error <- aov_tab["Residuals", "Mean Sq"]

b <- nlevels(datarbd$block)
t <- nlevels(datarbd$treatment)

RE <- ((b - 1) * MS_block + b * (t - 1) * MS_error) / ((b * t - 1) * MS_error)
RE
```

```
[1] 1.493642
```

Interpretation:

- **RE $>$ 1:** blocking improved precision. For example, RE = 1.5 means a CRD would need about 50% more replications to reach the same precision.
- **RE 1 or less:** blocking did not help much.

So hare the RBD increase the precision.

ANOVA only says that a difference exists somewhere, so a post-hoc comparison is needed.

---

5. Post-hoc Analysis

Post-hoc analysis is performed when the treatment effect is significant and we want to know **which treatment means differ**.

Here we use `emmeans` to obtain estimated marginal means (averaged over blocks) and compare treatments.

5.1 LSD comparison

LSD uses no multiplicity adjustment:

```r
library(emmeans)
```

```
Warning: package 'emmeans' was built under R version 4.4.3
```

```
Welcome to emmeans.
Caution: You lose important information if you filter this package's results.
See '? untidy'
```

```r
emm_rbd_lsd <- emmeans(modelrbd, pairwise ~ treatment, adjust = "none")
emm_rbd_lsd
```

```
$emmeans
 treatment    emmean     SE df lower.CL upper.CL
 Compost        4.00 0.0471 20     3.90     4.10
 Control        3.37 0.0471 20     3.27     3.47
 NPK            4.09 0.0471 20     3.99     4.18
 NPK_Organic    4.59 0.0471 20     4.49     4.69
 Organic        3.72 0.0471 20     3.62     3.82
 Urea           3.90 0.0471 20     3.80     3.99

Results are averaged over the levels of: block
Confidence level used: 0.95

$contrasts
 contrast               estimate     SE df t.ratio p.value
 Compost - Control         0.628 0.0666 20   9.436  <.0001
 Compost - NPK            -0.088 0.0666 20  -1.322  0.2010
 Compost - NPK_Organic    -0.592 0.0666 20  -8.895  <.0001
 Compost - Organic         0.278 0.0666 20   4.177  0.0005
 Compost - Urea            0.102 0.0666 20   1.533  0.1411
 Control - NPK            -0.716 0.0666 20 -10.758  <.0001
 Control - NPK_Organic    -1.220 0.0666 20 -18.330  <.0001
 Control - Organic        -0.350 0.0666 20  -5.259  <.0001
 Control - Urea           -0.526 0.0666 20  -7.903  <.0001
 NPK - NPK_Organic        -0.504 0.0666 20  -7.573  <.0001
 NPK - Organic             0.366 0.0666 20   5.499  <.0001
 NPK - Urea                0.190 0.0666 20   2.855  0.0098
 NPK_Organic - Organic     0.870 0.0666 20  13.072  <.0001
 NPK_Organic - Urea        0.694 0.0666 20  10.427  <.0001
 Organic - Urea           -0.176 0.0666 20  -2.644  0.0156

Results are averaged over the levels of: block
```

Compact letter display:

```r
library(multcomp)
```

```
Warning: package 'multcomp' was built under R version 4.4.3
```

```
Loading required package: mvtnorm
```

```
Loading required package: survival
```

```
Loading required package: TH.data
```

```
Loading required package: MASS
```

```

Attaching package: 'TH.data'
```

```
The following object is masked from 'package:MASS':

    geyser
```

```r
library(multcompView)
```

```
Warning: package 'multcompView' was built under R version 4.4.3
```

```r
multcomp::cld(emm_rbd_lsd$emmeans, adjust = "none", Letters = letters)
```

```
 treatment   emmean     SE df lower.CL upper.CL .group
 Control       3.37 0.0471 20     3.27     3.47  a
 Organic       3.72 0.0471 20     3.62     3.82   b
 Urea          3.90 0.0471 20     3.80     3.99    c
 Compost       4.00 0.0471 20     3.90     4.10    cd
 NPK           4.09 0.0471 20     3.99     4.18     d
 NPK_Organic   4.59 0.0471 20     4.49     4.69      e

Results are averaged over the levels of: block
Confidence level used: 0.95
significance level used: alpha = 0.05
NOTE: If two or more means share the same grouping symbol,
      then we cannot show them to be different.
      But we also did not show them to be the same.
```

Treatments sharing the same letter are **not statistically different**. Treatments with different letters are statistically different.

5.2 Tukey HSD comparison

Tukey's method controls the family-wise error rate across all pairwise comparisons.

```
emm_rbd_hsd <- emmeans(modelrbd, pairwise ~ treatment, adjust = "tukey")
emm_rbd_hsd
```

```
$emmeans
 treatment    emmean     SE df lower.CL upper.CL
 Compost        4.00 0.0471 20     3.90     4.10
 Control        3.37 0.0471 20     3.27     3.47
 NPK            4.09 0.0471 20     3.99     4.18
 NPK_Organic    4.59 0.0471 20     4.49     4.69
 Organic        3.72 0.0471 20     3.62     3.82
 Urea           3.90 0.0471 20     3.80     3.99

Results are averaged over the levels of: block
Confidence level used: 0.95

$contrasts
 contrast               estimate     SE df t.ratio p.value
 Compost - Control         0.628 0.0666 20   9.436  <.0001
 Compost - NPK            -0.088 0.0666 20  -1.322  0.7698
 Compost - NPK_Organic    -0.592 0.0666 20  -8.895  <.0001
 Compost - Organic         0.278 0.0666 20   4.177  0.0054
 Compost - Urea            0.102 0.0666 20   1.533  0.6487
 Control - NPK            -0.716 0.0666 20 -10.758  <.0001
 Control - NPK_Organic    -1.220 0.0666 20 -18.330  <.0001
 Control - Organic        -0.350 0.0666 20  -5.259  0.0005
 Control - Urea           -0.526 0.0666 20  -7.903  <.0001
 NPK - NPK_Organic        -0.504 0.0666 20  -7.573  <.0001
 NPK - Organic             0.366 0.0666 20   5.499  0.0003
 NPK - Urea                0.190 0.0666 20   2.855  0.0888
 NPK_Organic - Organic     0.870 0.0666 20  13.072  <.0001
 NPK_Organic - Urea        0.694 0.0666 20  10.427  <.0001
 Organic - Urea           -0.176 0.0666 20  -2.644  0.1320

Results are averaged over the levels of: block
P value adjustment: tukey method for comparing a family of 6 estimates
```

Compact letter display:

```r
multcomp::cld(emm_rbd_hsd$emmeans, adjust = "tukey", Letters = letters)
```

```
Note: adjust = "tukey" was changed to "sidak"
because "tukey" is only appropriate for one set of pairwise comparisons

 treatment   emmean     SE df lower.CL upper.CL .group
 Control       3.37 0.0471 20     3.23     3.51  a
 Organic       3.72 0.0471 20     3.58     3.86   b
 Urea          3.90 0.0471 20     3.76     4.03   bc
 Compost       4.00 0.0471 20     3.86     4.14    c
 NPK           4.09 0.0471 20     3.95     4.22    c
 NPK_Organic   4.59 0.0471 20     4.45     4.73     d

Results are averaged over the levels of: block
Confidence level used: 0.95
Conf-level adjustment: sidak method for 6 estimates
P value adjustment: tukey method for comparing a family of 6 estimates
significance level used: alpha = 0.05
NOTE: If two or more means share the same grouping symbol,
      then we cannot show them to be different.
      But we also did not show them to be the same.
```

LSD vs Tukey

- **LSD:** less conservative; more power, but higher Type I error risk across many comparisons.
- **Tukey:** more conservative; the safer default when all treatment pairs are compared.

---

6. Post-hoc Interpretation

Tukey's test ($= 0.05$) separated the treatments into four groups.

```r
NPK_Organic > NPK = Compost = Urea (bc) > Organic > Control
```

- `NPK_Organic` (4.59 t/ha) was significantly higher than all other treatments.
- `Control` (3.37 t/ha) was significantly lower than all other treatments.
- `NPK`, `Compost`, and `Urea` were not significantly different from each other.
- `Organic` was lower than `NPK` and `Compost`, but not different from `Urea`.

---

7. Agricultural Interpretation

`NPK_Organic` gave the highest yield (**4.59 t/ha**) and the Control the lowest (**3.37 t/ha**), a difference of about **1.22 t/ha**.

`NPK_Organic` also out-yielded `NPK` alone by about 0.50 t/ha. Cost, fertilizer rate, and economic return should be considered before making a recommendation.

Statistical significance alone does not mean that the top treatment is the best practical recommendation. Cost, fertilizer rate, economic return, and environmental effects should also be considered.